2.11 Graph Transformations: Combinations

Worked Solutions 1–41

1 a)

Stretch, parallel to the x-axis, scale factor $\frac{1}{4}$

Stretch, parallel to the y-axis, scale factor 3

(either order).

1 b)

Stretch, parallel to the x-axis, scale factor 3

Stretch, parallel to the y-axis, scale factor $\frac{1}{2}$

(either order).

2 a)

Translation by the vector $\begin{pmatrix} -6 \\ 0 \end{pmatrix}$

Reflection in the x-axis

(either order).

2 b)

Reflection in the y-axis

Translation by the vector $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$

(either order).

3 a)

Translation by the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Stretch, parallel to the y-axis, scale factor $\frac{1}{6}$

(either order).

3 b)

Stretch, parallel to the x-axis, scale factor $\frac{1}{8}$

Translation by the vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$

(either order).

4 a)

Translation by the vector $\begin{pmatrix} \frac{\pi}{3} \\ 0 \end{pmatrix}$

Stretch, parallel to the y-axis, scale factor 2

(either order).

4 b)

Stretch, parallel to the x-axis, scale factor 5

Translation by the vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$

(either order).

5 a)

Stretch, parallel to the x-axis, scale factor 3

Stretch, parallel to the y-axis, scale factor 5

(either order).

5 b)

Reflection in the y-axis

Translation by the vector $\begin{pmatrix} 0 \\ -9 \end{pmatrix}$

(either order).

6 a)

$$\text{Let } f(x) = \frac{1}{x}.$$
$$y = \frac{2}{x-4} = 2 \times \frac{1}{x-4} = 2f(x-4).$$

Translation by the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Stretch, parallel to the y-axis, scale factor 2

(either order).

6 b)

$$\text{Let } f(x) = \frac{1}{x}.$$
$$y = 5 - \frac{1}{x} = 5 + \frac{1}{-x} = 5 + f(-x).$$

Reflection in the y-axis

Translation by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$

(either order). **OR**

Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$.

7 a)

$$\text{Let } f(x) = \ln x.$$
$$y = \ln(2x) - 3 = f(2x) - 3.$$

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$

Translation by the vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

(either order).

7 b)

$$\text{Let } f(x) = \ln x.$$
$$y = \ln(2x) - 3 = \ln(2x) - \ln e^3 = \ln \frac{2x}{e^3}.$$

Stretch, parallel to the x-axis, scale factor $\frac{e^3}{2}$.

8 a)

$$(-6, 12) \mapsto (\frac{1}{3} \times -6, 12) = (-2, 12)$$
$$(-2, 12) \mapsto (-2, 2 \times 12) = (-2, 24).$$

Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$

followed by a

Stretch, parallel to the y-axis, scale factor 2.

8 b)

$$(-6, 12) \mapsto (-6 + 5, 12) = (-1, 12)$$
$$(-1, 12) \mapsto (-1, -1 \times 12) = (-1, -12).$$

Translation by the vector $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$

followed by a

Reflection in the x-axis.

9 a)

$$(4, -3) \mapsto (6 \times 4, -3) = (24, -3)$$
$$(24, -3) \mapsto (24, \frac{1}{3} \times -3) = (24, -1).$$

Stretch, parallel to the x-axis, scale factor 6

followed by a

Stretch, parallel to the y-axis, scale factor $\frac{1}{3}$.

9 b)

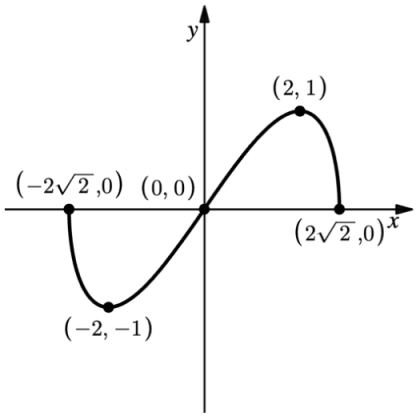
$$(4, -3) \mapsto (-1 \times 4, -3) = (-4, -3)$$
$$(-4, -3) \mapsto (-4, -3 + 9) = (-4, 6).$$

Reflection in the y-axis

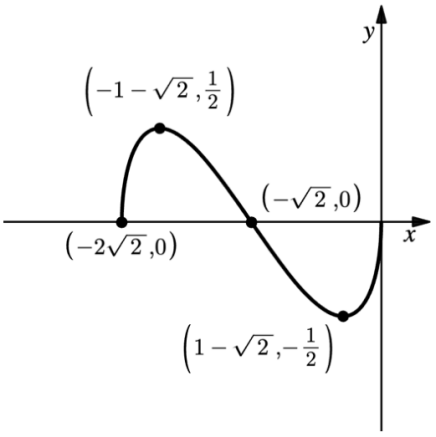
followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 9 \end{pmatrix}$.

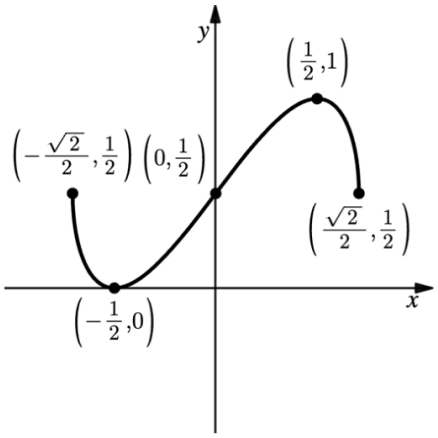
10 a)



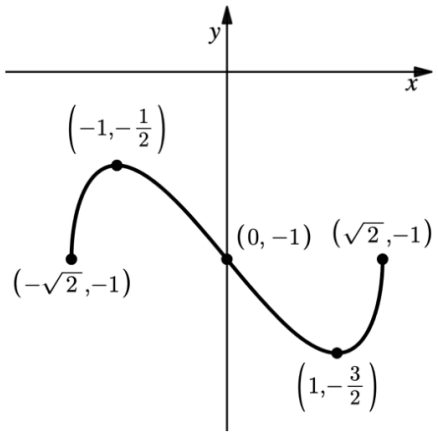
10 b)



10 c)



10 d)



11 a)

$$y = f(x) \mapsto y = 2f(x) \mapsto y = 2f(x) + 3.$$

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$.

11 b)

$$y = f(x) \mapsto y = 9f(x) \mapsto y = 9f(x) - 8.$$

Stretch, parallel to the y-axis, scale factor 9

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$.

11 c)

$$y = f(x) \mapsto y = \frac{1}{5}f(x) \mapsto y = \frac{1}{5}f(x) - 10.$$

Stretch, parallel to the y-axis, scale factor $\frac{1}{5}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -10 \end{pmatrix}$.

11 d)

$$y = f(x) \mapsto y = \frac{3}{2}f(x) \mapsto y = \frac{3}{2}f(x) + 20.$$

Stretch, parallel to the y-axis, scale factor $\frac{3}{2}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 20 \end{pmatrix}$.

12 a)

$$\begin{aligned} y = 4f(x) - 1 \mapsto \frac{4}{5}y &= 4f(x) - 1 \\ \Rightarrow y &= 5f(x) - \frac{5}{4} \\ \Rightarrow y - \frac{13}{4} &= 5f(x) - \frac{5}{4} \\ \Rightarrow y &= 5f(x) + 2. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor $\frac{5}{4}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ \frac{13}{4} \end{pmatrix}$.

12 b)

$$\begin{aligned} y = -f(x) - 5 \mapsto 2y &= -f(x) - 5 \\ \Rightarrow y &= -\frac{1}{2}f(x) - \frac{5}{2} \\ \Rightarrow y - \frac{19}{2} &= -\frac{1}{2}f(x) - \frac{5}{2} \\ \Rightarrow y &= -\frac{1}{2}f(x) + 7. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor $\frac{1}{2}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ \frac{19}{2} \end{pmatrix}$.

13 a)

$$y = f(x) \mapsto y = f(x+1) \mapsto y = f(3x+1).$$

Translation by the vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$.

13 b)

$$y = f(x) \mapsto y = f(x-9) \mapsto y = f(2x-9).$$

Translation by the vector $\begin{pmatrix} 9 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

13 c)

$$y = f(x) \mapsto y = f(x+2) \mapsto y = f(-x+2) = f(2-x).$$

Translation by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$

followed by a

Reflection in the y-axis.

13 d)

$$y = f(x) \mapsto y = f(x-3) \mapsto y = f(-x-3).$$

Translation by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$

followed by a

Reflection in the y-axis.

14 a)

$$\begin{aligned} y = f(3x+8) &\mapsto y = f(3(x-4)+8) = f(3x-4) \\ &\mapsto y = f(3(2x)-4) = f(6x-4). \end{aligned}$$

Translation by the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

14 b)

$$y = f(2-x) \mapsto y = f(2-(x-1)) = f(3-x)$$
$$\mapsto y = f(3-\frac{x}{5}).$$

Translation by the vector $\begin{pmatrix}1\\0\end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 5.

15 a)

$$(10,-3) \mapsto (10, 2\times -3) = (10,-6)$$
$$(10,-6) \mapsto (10+5, -6+3) = (15,-3).$$

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix}5\\3\end{pmatrix}$.

15 b)

$$(10,-3) \mapsto (10, -1\times -3) = (10,3)$$
$$(10,3) \mapsto (10, 3-9) = (10,-6).$$

Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix}0\\-9\end{pmatrix}$.

15 c)

$$(10,-3) \mapsto (10-5, -3) = (5,-3)$$
$$(5,-3) \mapsto (\frac{1}{3}\times 5, -3) = (\frac{5}{3}, -3).$$

Translation by the vector $\begin{pmatrix}-5\\0\end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$.

15 d)

$$(10, -3) \mapsto (10 + 2, -3) = (12, -3)$$
$$(12, -3) \mapsto (4 \times 12, -3) = (48, -3).$$

Translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 4.

16 a)

Stretch, parallel to the y-axis, scale factor 4

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -6 \end{pmatrix}$.

16 b)

Translation by the vector $\begin{pmatrix} \frac{\pi}{9} \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 6.

16 c)

$$y = \sin x \mapsto y = \sin(x + \frac{\pi}{2}) = \cos x \mapsto y = \cos 2x.$$

Translation by the vector $\begin{pmatrix} -\frac{\pi}{2} \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

16 d)

$$y = \sin x \mapsto y = \sin(x + \frac{\pi}{2}) = \cos x \mapsto y = 3\cos x.$$

Translation by the vector $\begin{pmatrix} -\frac{\pi}{2} \\ 0 \end{pmatrix}$

Stretch, parallel to the y-axis, scale factor 3

(either order).

17 a)

Translation by the vector $\begin{pmatrix} 10 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{6}$.

17 b)

Translation by the vector $\begin{pmatrix} -9 \\ 0 \end{pmatrix}$

followed by a

Reflection in the y-axis.

17 c)

Stretch, parallel to the y-axis, scale factor 10

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 20 \end{pmatrix}$.

17 d)

Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 18 \end{pmatrix}$.

18 a)

$$y = f(x) \mapsto y = f(5x - 2).$$

Translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{5}$.

18 b)

$$y = f(x) \mapsto y = 8f(x) - 9.$$

Stretch, parallel to the y-axis, scale factor 8

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -9 \end{pmatrix}$.

18 c)

$$y = \frac{6 - 5x}{x} = \frac{6}{x} - \frac{5x}{x} = \frac{6}{x} - 5$$
$$y = f(x) \mapsto y = 6f(x) - 5.$$

Stretch, parallel to the y-axis, scale factor 6

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$.

18 d)

$$y = \frac{2x+7}{20x} = \frac{2x}{20x} + \frac{7}{20x} = \frac{1}{10} + \frac{7}{20x}$$
$$y = f(x) \mapsto y = \frac{1}{10} + \frac{7}{20}f(x).$$

Stretch, parallel to the y-axis, scale factor $\frac{7}{20}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ \frac{1}{10} \end{pmatrix}$.

19 a)

$$y = f(x) \mapsto y = -\frac{1}{5}f(x).$$

Stretch, parallel to the y-axis, scale factor $\frac{1}{5}$

Reflection in the x-axis

(either order).

19 b)

$$y = f(x) \mapsto y = 1 + 2f(x).$$

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

19 c)

$$y = \frac{2+3x^2}{6x^2} = \frac{2}{6x^2} + \frac{3x^2}{6x^2} = \frac{1}{3x^2} + \frac{1}{2}$$
$$y = f(x) \mapsto y = \frac{1}{3}f(x) + \frac{1}{2}.$$

Stretch, parallel to the y-axis, scale factor $\frac{1}{3}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}$.

19 d)

$$y = \frac{1-6x^2}{8x^2} = \frac{1}{8x^2} - \frac{6x^2}{8x^2} = \frac{1}{8x^2} - \frac{3}{4}$$
$$y = f(x) \mapsto y = \frac{1}{8}f(x) - \frac{3}{4}.$$

Stretch, parallel to the y-axis, scale factor $\frac{1}{8}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -\frac{3}{4} \end{pmatrix}$.

20 a)

$$y = \ln(x+1) \mapsto y = \ln((x+2)+1) = \ln(x+3)$$
$$\mapsto y = \ln(3x+3).$$

Translation by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$.

20 b)

$$y = \ln(x+1) \mapsto y - \ln 3 = \ln(x+1)$$
$$\Rightarrow y = \ln 3 + \ln(x+1) = \ln(3(x+1)) = \ln(3x+3).$$

Translation by the vector $\begin{pmatrix} 0 \\ \ln 3 \end{pmatrix}$.

21 a)

$$y = 4x+3 \mapsto y = 4(x+1)+3 \Rightarrow y = 4x+7$$
$$\mapsto y = 4(2x)+7 = 8x+7.$$

Translation by the vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

21 b)

$$y = 4x+3 \mapsto y = 4(x+\frac{3}{4})+3 \Rightarrow y = 4x+6$$
$$\mapsto y = 4(\frac{x}{2})+6 = 2x+6.$$

Translation by the vector $\begin{pmatrix} -\frac{3}{4} \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 2.

22 a)

$$y = 4-5x \mapsto y = 4-5(x+\frac{12}{5}) \Rightarrow y = -8-5x$$
$$\mapsto y = -8-5(2x) = -10x-8.$$

Translation by the vector $\begin{pmatrix} -\frac{12}{5} \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

22 b)

$$y = 4 - 5x \mapsto -y = 4 - 5x \Rightarrow y = 5x - 4$$
$$\mapsto \frac{1}{10}y = 5x - 4 \Rightarrow y = 50x - 40.$$

Reflection in the x-axis

Stretch, parallel to the y-axis, scale factor 10

(either order).

23)

$$y = f(x) \mapsto y = f(x - 2) - 3$$
$$y = 5x + 2 \mapsto y = 5(x - 2) + 2 - 3 \Rightarrow y = 5x - 11$$
$$y = f(x - 2) - 3 \mapsto y = f(\frac{x}{2} - 2) - 3$$
$$y = 5x - 11 \mapsto y = 5(\frac{x}{2}) - 11 \Rightarrow y = \frac{5}{2}x - 11.$$

24)

$$y = f(x) \mapsto y = \frac{1}{3}f(x)$$
$$y = 8 - 6x \mapsto y = \frac{1}{3}(8 - 6x) \Rightarrow y = \frac{8}{3} - 2x$$
$$y = \frac{1}{3}f(x) \mapsto y = \frac{1}{3}f(x + \frac{2}{3}) + \frac{4}{3}$$
$$y = \frac{8}{3} - 2x \mapsto y = \frac{8}{3} - 2(x + \frac{2}{3}) + \frac{4}{3} \Rightarrow y = \frac{8}{3} - 2x.$$

25 a)

Stretch, parallel to the y-axis, scale factor 4

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 9 \end{pmatrix}$.

25 b)

Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$.

25 c)

Stretch, parallel to the y-axis, scale factor $\frac{1}{4}$

followed by a

Translation by the vector $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$.

25 d)

Stretch, parallel to the y-axis, scale factor 8

followed by a

Translation by the vector $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$.

25 e)

Translation by the vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{4}$.

25 f)

Translation by the vector $\begin{pmatrix} 9 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 3.

25 g)

$$\begin{aligned} y &= 2x^2 - 4x - 7 = 2(x^2 - 2x) - 7 \\ &= 2((x-1)^2 - 1) - 7 \\ &= 2(x-1)^2 - 2 - 7 = 2(x-1)^2 - 9. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix} 1 \\ -9 \end{pmatrix}$.

25 h)

$$\begin{aligned} y &= 3x^2 + 12x + 1 = 3(x^2 + 4x) + 1 \\ &= 3((x+2)^2 - 4) + 1 \\ &= 3(x+2)^2 - 12 + 1 = 3(x+2)^2 - 11. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor 3

followed by a

Translation by the vector $\begin{pmatrix} -2 \\ -11 \end{pmatrix}$.

25 i)

$$\begin{aligned} y &= -(x+1)(x-9) = -(x^2 - 8x - 9) \\ &= -((x-4)^2 - 16 - 9) = -(x-4)^2 + 25. \end{aligned}$$

Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 4 \\ 25 \end{pmatrix}$.

25 j)

$$\begin{aligned} y &= (6x-4)(3x+10) = 18x^2 + 48x - 40 \\ &= 18(x^2 + \frac{8}{3}x) - 40 \\ &= 18((x + \frac{4}{3})^2 - \frac{16}{9}) - 40 = 18(x + \frac{4}{3})^2 - 72. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor 18

followed by a

Translation by the vector $\begin{pmatrix} -\frac{4}{3} \\ -72 \end{pmatrix}$.

26 a)

$$\begin{aligned} y = x^2 + 2 \mapsto \frac{1}{9}y = x^2 + 2 &\Rightarrow y = 9x^2 + 18 \\ \mapsto y + 8 = 9x^2 + 18 &\Rightarrow y = 9x^2 + 10. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor 9

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$.

26 b)

$$\begin{aligned} y = x^2 + 2 \mapsto \frac{1}{8}y = x^2 + 2 &\Rightarrow y = 8x^2 + 16 \\ \mapsto y + 16 = 8x^2 + 16 &\Rightarrow y = 8x^2. \end{aligned}$$

Stretch, parallel to the y-axis, scale factor 8

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -16 \end{pmatrix}$.

26 c)

$$y = x^2 + 2 \mapsto -y = x^2 + 2 \Rightarrow y = -x^2 - 2$$
$$\mapsto y - 9 = -x^2 - 2 \Rightarrow y = 7 - x^2.$$

Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 9 \end{pmatrix}$.

27)

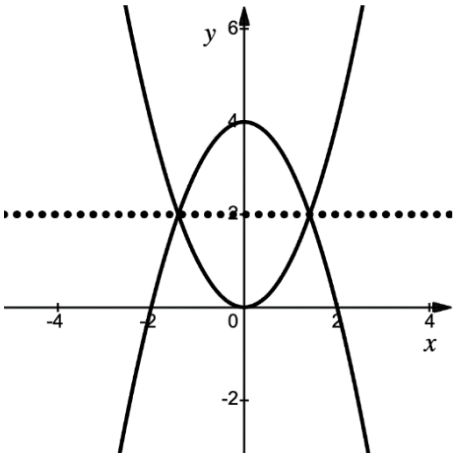
$$y = x^3 \mapsto y = (2x)^3 \Rightarrow y = 8x^3$$
$$\mapsto y = 8(x+1)^3 + 2 \quad (\Rightarrow y = 8x^3 + 24x^2 + 24x + 10).$$

28)

$$y = \ln x \mapsto y = \ln \frac{x}{9}$$
$$y = \ln \frac{x}{9} \mapsto y = \ln \frac{x-6}{9} - \ln 3$$
$$\Rightarrow y = \ln \frac{(x-6)/9}{3} \Rightarrow y = \ln(\frac{x}{27} - \frac{2}{9}).$$

29 a)

$$y = x^2 \mapsto -y = x^2 \mapsto -(y-4) = x^2$$
$$\Rightarrow y - 4 = -x^2 \Rightarrow y = 4 - x^2.$$



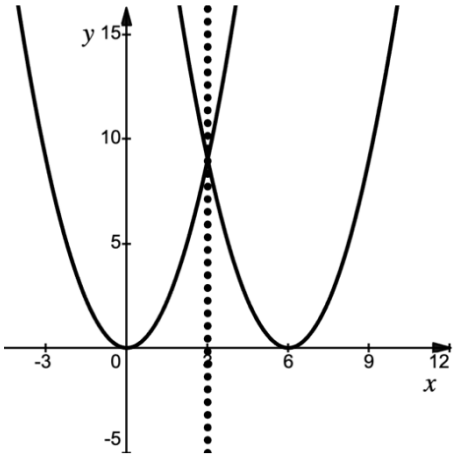
Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$.

29 b)

$$y = x^2 \mapsto y = (-x)^2 \mapsto y = (-(x-6))^2$$
$$\Rightarrow y = (x-6)^2.$$



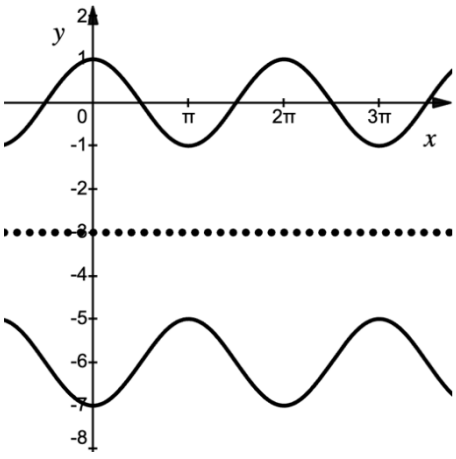
Reflection in the y-axis

followed by a

Translation by the vector $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$.

30 a)

$$y = \cos x \mapsto -y = \cos x \mapsto -(y+6) = \cos x$$
$$\Rightarrow y + 6 = -\cos x \Rightarrow y = -\cos x - 6.$$



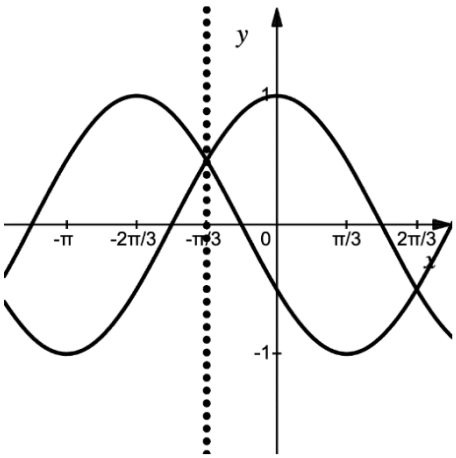
Reflection in the x-axis

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -6 \end{pmatrix}$.

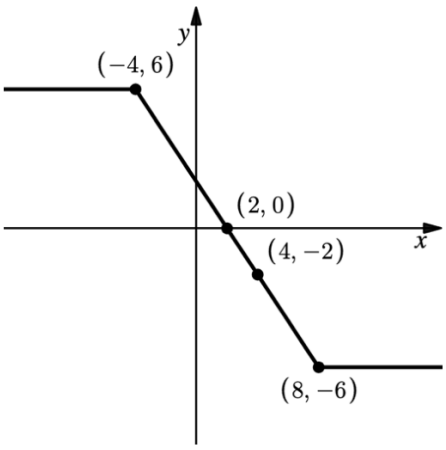
30 b)

$$y = \cos x \mapsto y = \cos(-x) \mapsto y = \cos(-(x + \frac{2\pi}{3}))$$
$$\Rightarrow y = \cos(x + \frac{2\pi}{3}).$$

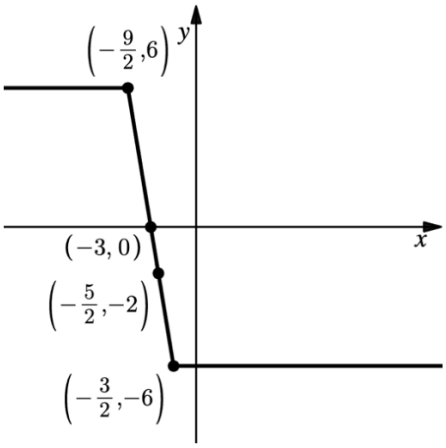


Reflection in the y-axis
followed by a
Translation by the vector $\begin{pmatrix} -\frac{2\pi}{3} \\ 0 \end{pmatrix}$.

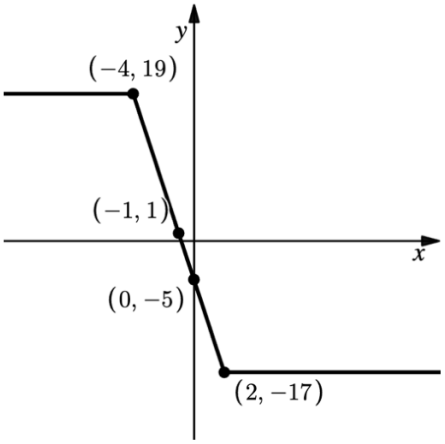
31 a)



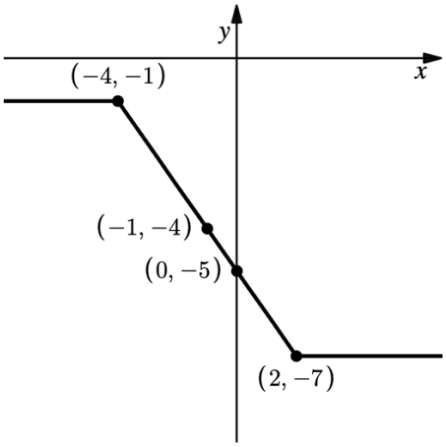
31 b)


$\left(-\frac{9}{2},6\right)$
y
x
$(-3,0)$
$\left(-\frac{5}{2},-2\right)$
$\left(-\frac{3}{2},-6\right)$

31 c)


y
$(-4,19)$
$(-1,1)$
x
$(0,-5)$
$(2,-17)$

31 d)


y
x
$(-4,-1)$
$(-1,-4)$
$(0,-5)$
$(2,-7)$

32)

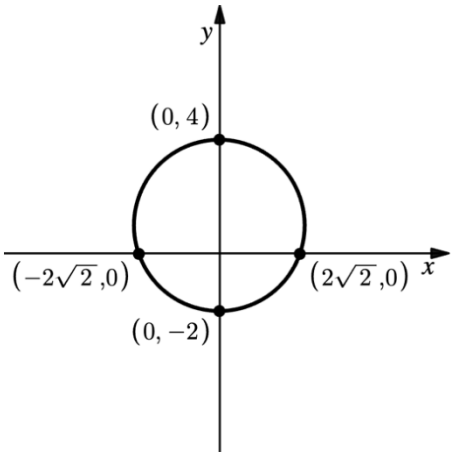
Reflection in the line $y = x$.

33)

Reflection in the line $y = x$.

34)

$$(y-1)^2 + x^2 = 9.$$



35)

$$y = \ln x \mapsto y = e^x \mapsto y = e^{3x}.$$

Reflection in the line $y = x$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$.

36)

$$y = \ln(x-5) \mapsto y = \ln((x+5)-5)$$
$$\Rightarrow y = \ln x \mapsto y = e^x.$$

Translation by the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$

followed by a

Reflection in the line $y = x$.

37)

$$y = \ln x \mapsto y = e^x$$
$$\Rightarrow y = e^x \mapsto \frac{1}{2}y = e^x \Rightarrow y = 2e^x$$
$$\Rightarrow y = 2e^x \mapsto y + 7 = 2e^x \Rightarrow y = 2e^x - 7.$$

Reflection in the line $y = x$

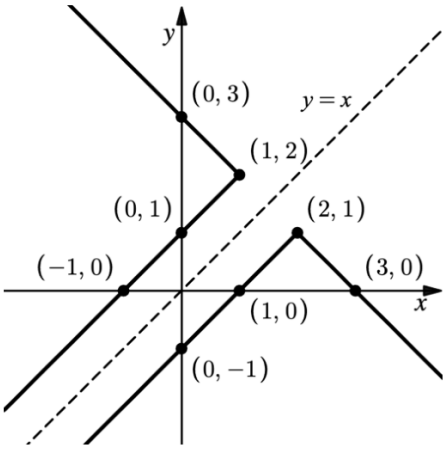
followed by a

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -7 \end{pmatrix}$.

38)



39)

Translation by the vector $\begin{pmatrix} 9 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 3

followed by a

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$.

Alternatively:

Stretch, parallel to the y-axis, scale factor 2

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

followed by a

Translation by the vector $\begin{pmatrix} 9 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor 3.

40)

Translation by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{5}{2}$

followed by a

Stretch, parallel to the y-axis, scale factor $\frac{3}{5}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

Alternatively:

Stretch, parallel to the y-axis, scale factor $\frac{3}{5}$

followed by a

Translation by the vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$

followed by a

Translation by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

followed by a

Stretch, parallel to the x-axis, scale factor $\frac{5}{2}$.

41)

Consider reflecting in the line $y = x$ first (this reverses the orientation of the coordinates), and then reflect in the y-axis:

$$y = x^2 \mapsto x = y^2 \mapsto -x = y^2.$$

